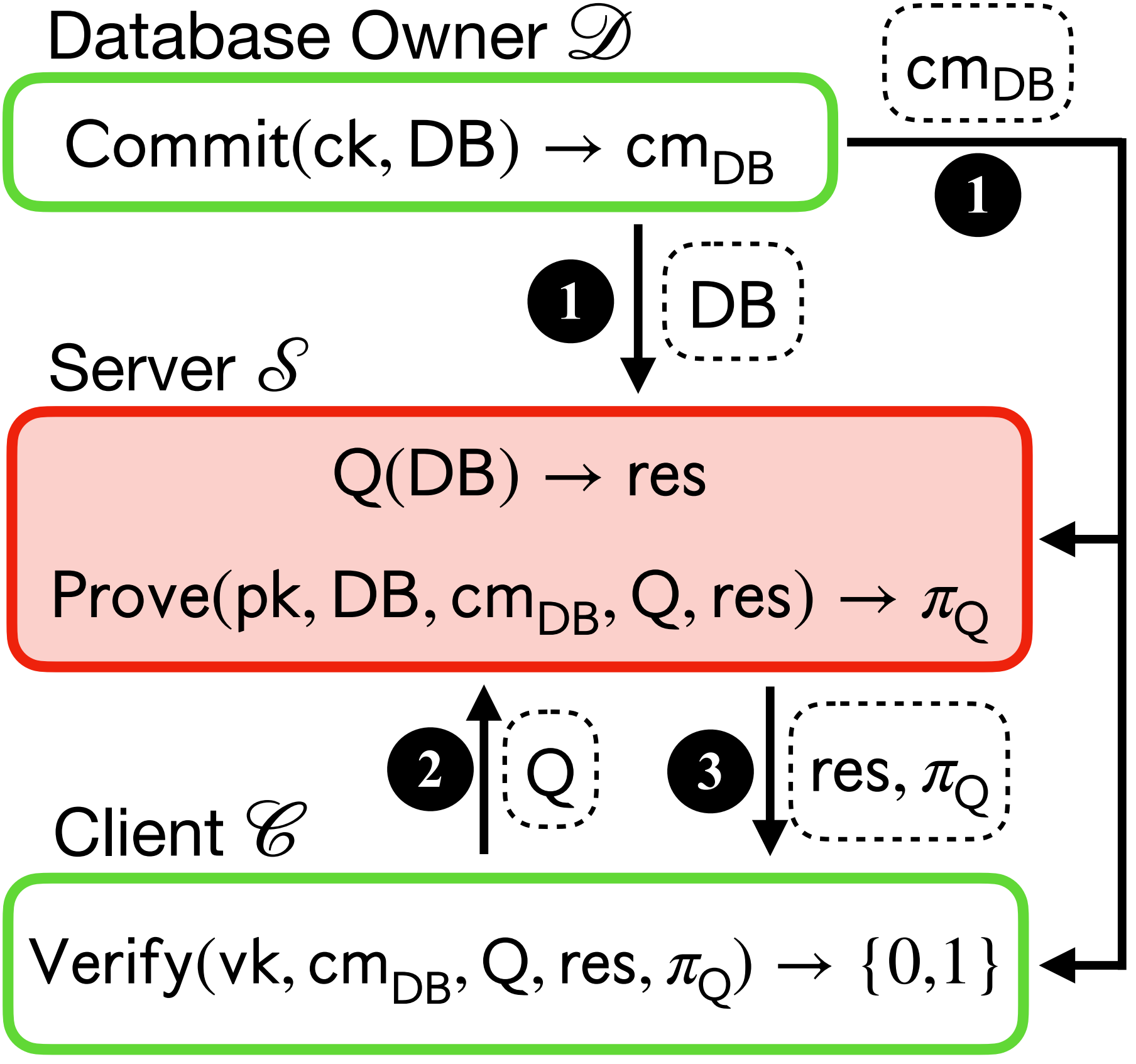
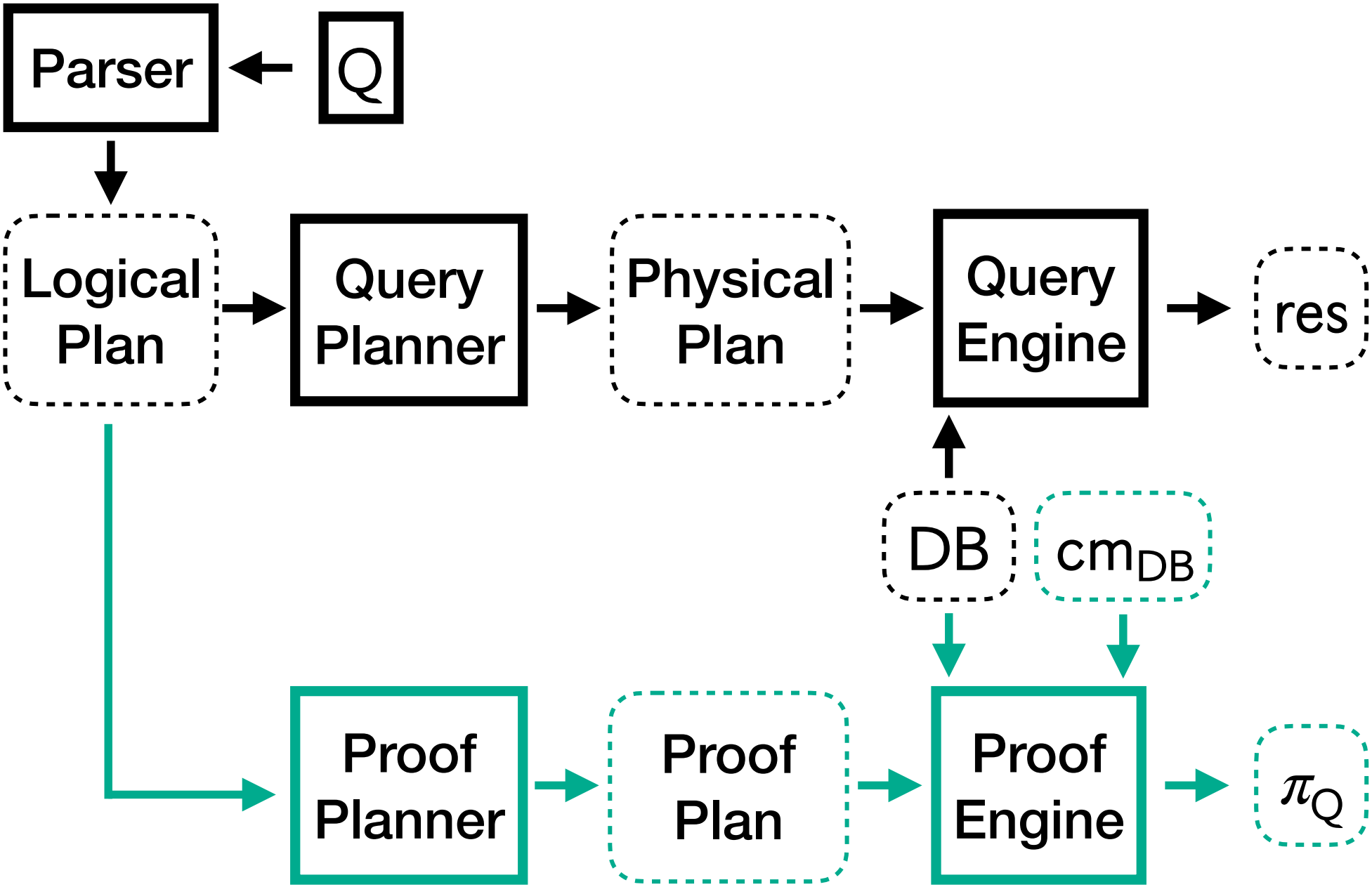
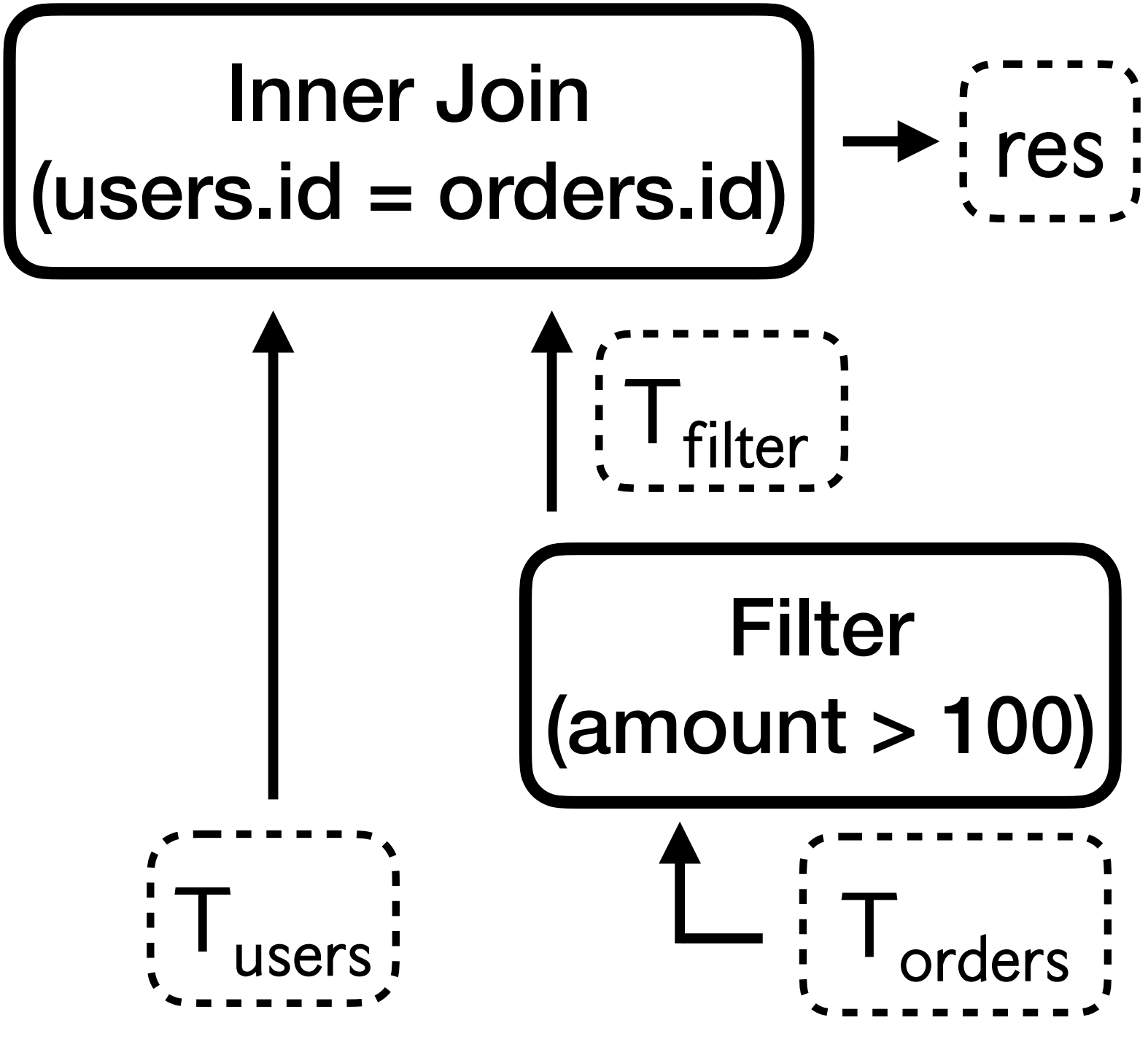
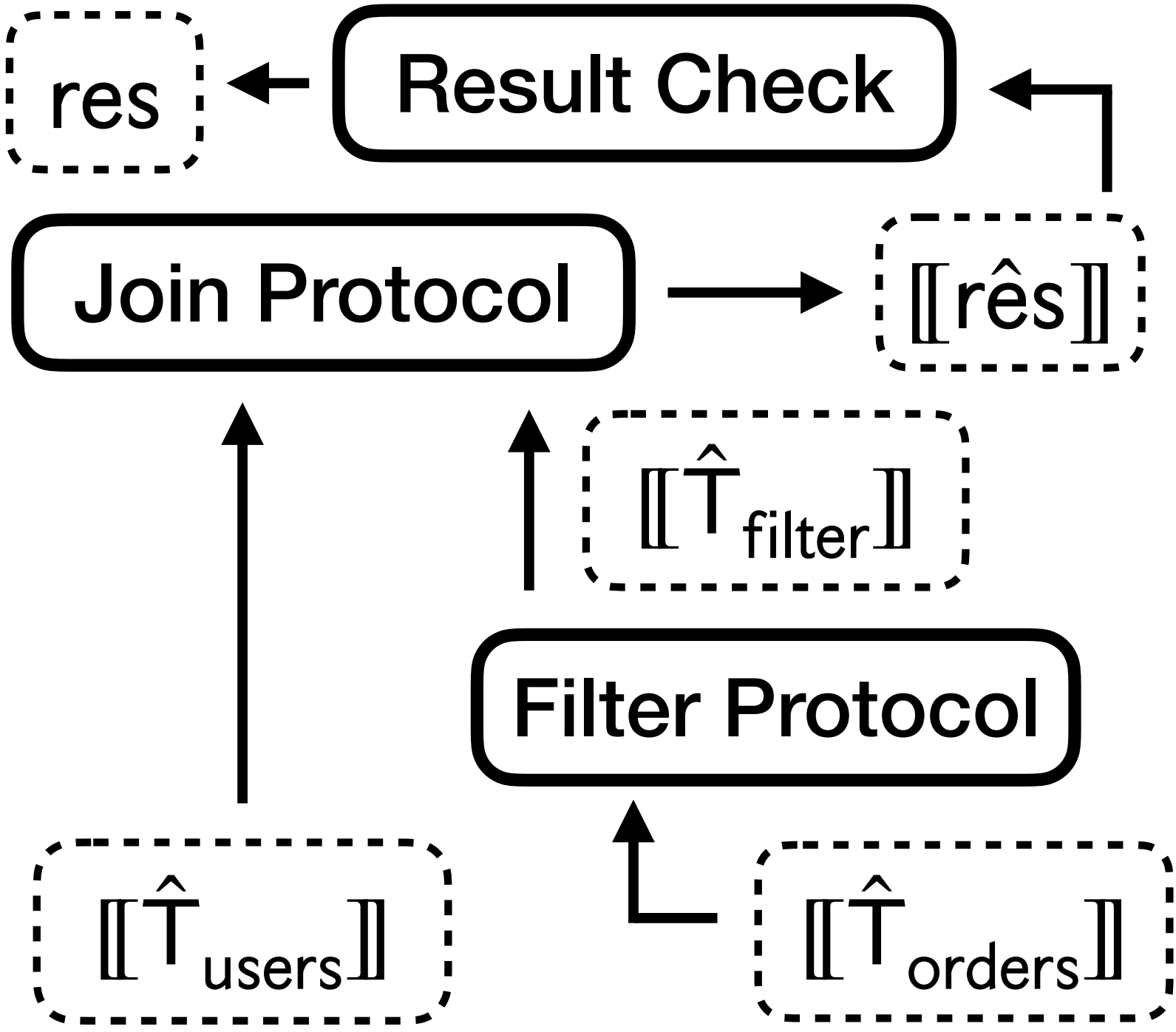
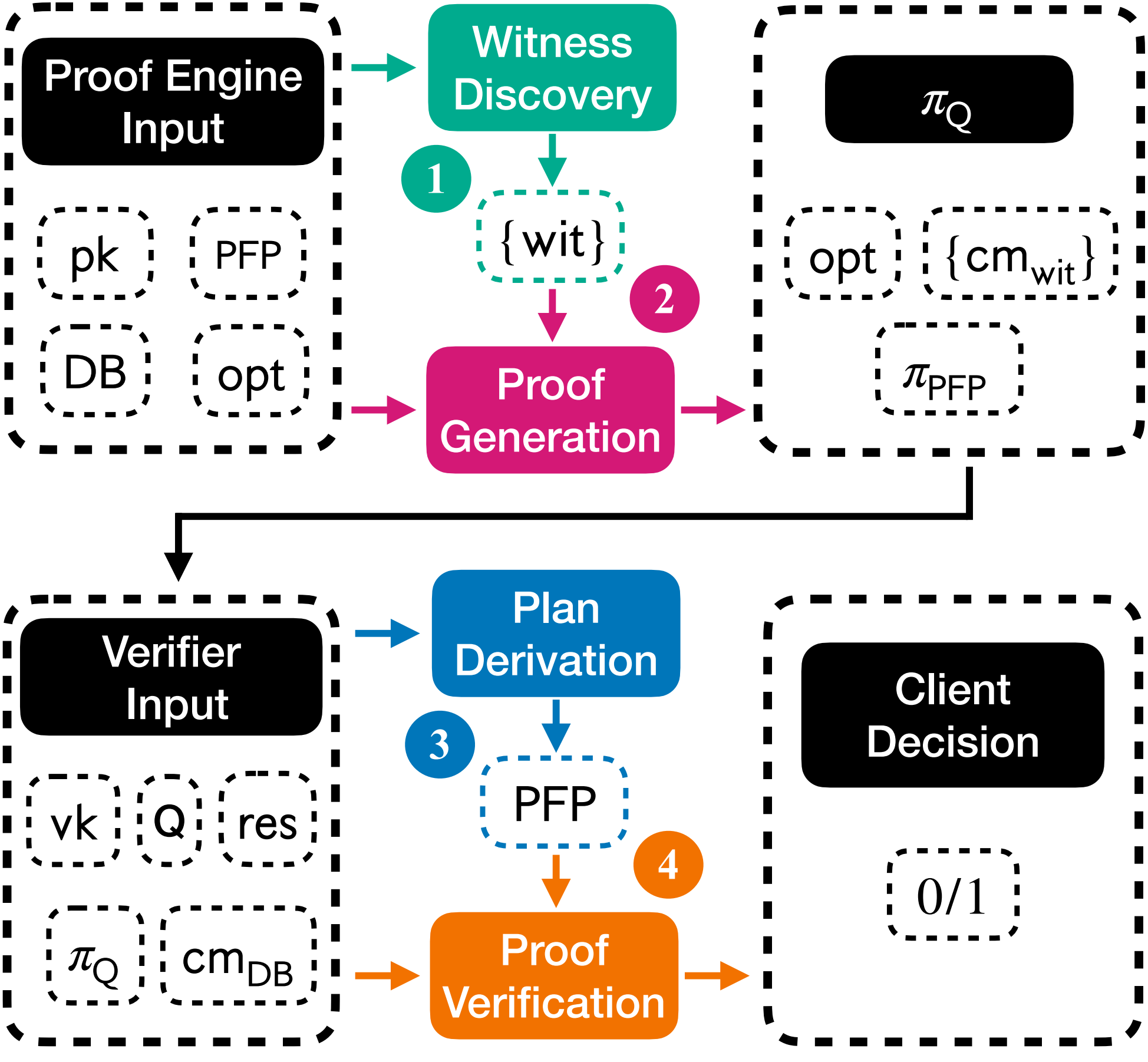

Database Owner $\mathscr{D}$
Commit(ck, DB) → $\mathsf{cm}_{\mathsf{DB}}$
$\mathsf{cm}_{\mathsf{DB}}$
1
1
DB
Server $\mathscr{S}$
Q(DB) → res
Prove(pk, DB, $\mathsf{cm}_{\mathsf{DB}}$, Q, res) → π_{Q}
2
Q
3
res, π_{Q}
Client $\mathscr{C}$
Verify(vk, $\mathsf{cm}_{\mathsf{DB}}$, Q, res, π_{Q}) → {0,1}


Parser
Q
Logical Plan
Query Planner
Physical Plan
Query Engine
res
DB
cm_{DB}
Proof Planner
Proof Plan
Proof Engine
π_Q


Inner Join
(users.id = orders.id)
res
T filter
Filter
(amount > 100)
T users
T orders


res
Result Check
Join Protocol
[[rês]]
[[T̂ filter]]
Filter Protocol
[[T̂ users]]
[[T̂ orders]]


Proof Engine Input
pk
PFP
DB
opt
Witness Discovery
1
{wit}
2
Proof Generation
π_Q
opt
$\{cm_{wit}\}$
π_{PFP}
Verifier Input
vk
Q
res
π_Q
cm_{DB}
Plan Derivation
3
PFP
4
Proof Verification
Client Decision
0/1